

Sources and Implications of Labor Market Informality: The Case of Minimum Wage Policies

(Preliminary and incomplete)

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Conference – Venue TBD – 2026

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Introduction

Minimum wage policies are among the most widely used labor market interventions in the world, yet most of what we know comes from **high-income economies with low informality**.

- In Latin America and the Caribbean (LAC), informal employment routinely accounts for **half or more of the labor force**:
 - a binding minimum wage governs, by law, **only formal salaried jobs**;
 - but workers move fluidly across unemployment, formal salaried work, informal salaried work, and self-employment.
- So a minimum wage increase does **not** simply raise wages or destroy jobs at the margin. It can:
 - reshape the **composition of employment** across sectors;
 - push workers from regulated into unregulated jobs;
 - propagate to wages it does not legally cover.
- Standard tools miss most of this:
 - wage distributions and employment rates ignore the equilibrium reallocation of workers *and firms* across sectors;
 - they are silent about workers who **transit repeatedly** across states over a career.

We develop and estimate a **search and matching model** built for exactly this environment.

Model:

- Frictional matching, match-specific productivity, wages set by **Nash bargaining**.
- Four states: unemployment, formal salaried, informal salaried, **(informal) self-employment**.
- Institutions enter explicitly: payroll contribution and benefit on formal jobs, punishment cost on informal jobs, and a minimum wage as a **floor on formal bargaining**.
- Firms choose **endogenously** whether to post a formal or an informal vacancy, so the composition of labor demand responds to policy.

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Estimation — Colombia:

- Maximum likelihood on the joint distribution of **states, durations, and accepted wages**.
- GEIH 2023, young (18–30) workers with at most high school: the segment **most exposed** to the minimum wage.
- Plus **job-posting counts** from the IDB Labor Observatory, to recover the demand side.

We contribute to two strands of the empirical search and matching literature:

- Informality: [Albrecht et al. (2009); Bosch and Esteban-Pretel (2012); Meghir et al. (2015); Bobba et al. (2021, 2022)]
- Minimum wage policy: [Flinn (2006); Eckstein et al. (2011); Flinn and Mullins (2021); Flabbi (2021)]

We advance both along three dimensions:

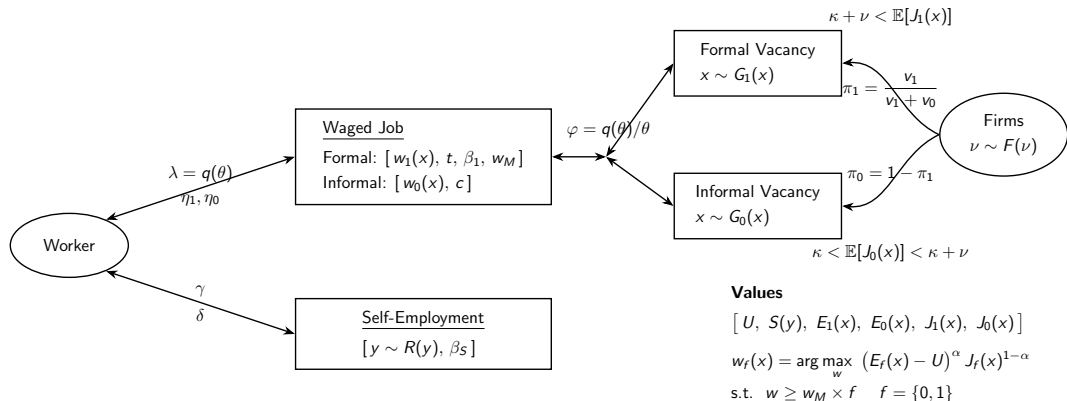
1. We model the minimum wage as a **binding constraint on bargaining** and let it propagate through workers' outside options to the uncovered informal sector:
 - the **lighthouse effect** emerges as an equilibrium outcome rather than being imposed.
2. We endogenize the formal/informal composition of **labor demand** through firms' vacancy-posting choices:
 - this margin turns out to be **quantitatively decisive**, and it is shut down when arrival rates are held fixed, as in most prior work.
3. We treat **self-employment** as a distinct state with its own productivity, mobility and amenity value, rather than folding it into informal salaried work or unemployment.

Preview of Preliminary Results

1. The model fits the Colombian labor market closely, including the two features that motivate it:
 - high informality (only 64% of salaried employees are formal) and a strongly binding minimum wage (about 50% of formal employees earn exactly w_M).
2. Self-employment carries a positive flow value — about 22% of the minimum wage, and even larger than the estimated value of formal employment:
 - consistent with valued amenities, not with mere disguised unemployment.
3. The impact of the minimum wage on worker welfare is non-monotonic:
 - a moderate increase raises welfare; increases beyond roughly the current level lower it sharply, pushing workers out of formal jobs into informal work while aggregate production falls;
 - the welfare-maximizing minimum wage is positive and close to current levels.
4. These effects are driven by firms' endogenous vacancy posting:
 - the employment response is an order of magnitude larger than when vacancy creation is held fixed;
 - the amenity value of self-employment cushions the adjustment.

Model

Search and Matching + Match Specific Productivity + Wage Bargaining + Endogenous Vacancy Posting + Minimum Wage + Informality



Model: Empirical Implications — Five Facts

The model reproduces, **simultaneously**, the five regularities of high-informality labor markets:

1. A significant proportion of **ex-ante identical workers in jobs of different formality status**.
 - Unemployed workers optimally take whichever of the three employment states the draw of x or y makes best.
2. **Two-way transitions** between formal and informal jobs.
 - Every job ends in unemployment, from which any transition is possible.
3. Higher average formal wages, but with **substantial overlap** between the two wage distributions.
 - Delivered by $G_1(x)$ stochastically dominating $G_0(x)$ with enough dispersion.
4. A **spike at exactly the minimum wage** in the formal wage distribution.
 - All meetings with $\underline{x}_M \leq x < \bar{x}_M$ are accepted and pay exactly w_M .
5. Informal wages that respond to the minimum wage: the **lighthouse effect**.
 - $w_0(x)$ depends on ρU , which depends on w_M . A worker bargaining over an informal job knows he could return to search and find a formal job subject to the floor.

Data

Supply side — Gran Encuesta Integrada de Hogares (GEIH) 2023:

- Labor market state, **ongoing duration** in it, and wages/earnings. Formality by the standard ILO definition: currently contributing to social security.

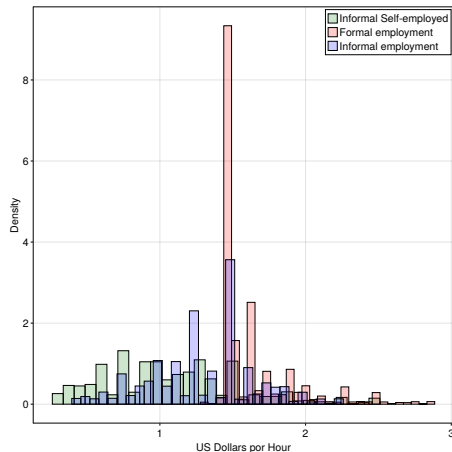
Estimation sample: full-time (at least 48 h/week), **18–30 years old, at most high school**

- the segment most exposed to the minimum wage, and homogeneous enough for the model's ex-ante homogeneity assumption. $N = 21,409$.
- States: 25.4% unemployed, 49.3% salaried employees (of whom **64% formal**), 25.3% self-employed.
- Hourly wages: formal US\$1.6, informal US\$1.3 (a **23% gap**), self-employed US\$1.0, the last with the widest dispersion (SD 0.46 vs 0.28, 0.37).
- Durations: 5 months unemployed; 1.8 years formal; 1.5 years informal; **3 years self-employed**.

Demand side — IDB Labor Observatory:

- Google Jobs (formal) postings requiring **little or no skill**: 12,378, against 1,501,100 unemployed aged 18–30 with at most high school $\Rightarrow$ formal market tightness $\theta = 0.0082$.

Figure 1: Self-employed Income and Wage Distributions by Labor Market State



- The monthly minimum wage is 1,160,000 pesos (US\$1.456 per hour).
- The spike accounts for 44% of full-time wage earners (3,249 observations) — and about half of formal employees.

Estimation

Supply side — maximum likelihood. Each individual contributes the joint probability of her state, her ongoing duration, and her wage/earnings likelihood

- State probabilities come from the **steady-state flow balance**; durations are exponential with the state's hazard.
- Accepted wage densities are G_f / R mapped through the wage schedules and **truncated at the reservation values**.
- Formal employees split into a **mass point at w_M** and a continuous part above it: the spike is an object *in* the likelihood, not an add-on.
- G_1, G_0, R lognormal; multiplicative lognormal measurement error with $E[\epsilon|w] = 1 \Rightarrow \mu_\epsilon = -\sigma_\epsilon^2/2$.

Demand side — recovered from the supply-side estimates plus the vacancy count equations

- π_1 from $\lambda_1/(\lambda_1 + \lambda_0)$; matching elasticity ϕ from $\lambda = \theta^\phi$; posting cost κ from informal free entry; ψ from $\pi_1 = F(\nu^*)$ with ν exponential.

Calibrated: $t = 0.16$ [OECD]; $\rho = 0.10$ [Moore et al. (2020)]; $c = 0.13$, i.e. 13% of the average informal wage [Bobba et al. (2022)]; $\alpha = 0.5$ (no firm-side data); $w_M = 1.456$; $\theta = 0.0082$.

Table 1: Parameter Estimates

Parameter	Estimate	Std. Error	Parameter	Estimate	Std. Error
Dynamics			Utility		
λ_1	0.078	(0.0058)	β_u	-3.542	(23.5913)
λ_0	0.050	(0.0030)	β_1	0.303	(17.0538)
γ	0.033	(0.0017)	β_s	0.321	(0.1902)
η_1	0.049	(0.0008)	Measurement Error		
η_0	0.056	(0.0013)	σ_ϵ	0.043	(0.0247)
δ	0.027	(0.0004)	Demand Side		
Productivity Distribution			ϕ	0.429	(0.0115)
μ_y	-0.069	(0.0532)	π_1	0.610	(0.0168)
σ_y	0.462	(0.0133)	κ	234.1	(41.333)
μ_1	1.202	(0.6065)	ψ	0.005	(0.0080)
σ_1	0.255	(0.0786)	<i>LogL</i> 53847.0		
μ_0	0.815	(0.0808)			
σ_0	0.349	(0.0288)			

NOTE: Maximum likelihood; bootstrap standard errors in parentheses. Demand-side parameters recovered from the ML estimates as described in the previous slide. Conditioning on calibrated $w_M = 1.456$, $t = 0.16$, $c = 0.13$, $\alpha = 0.5$, $\theta = 0.0082$.

Table 2: Implied Estimated Values

Definition	Value	Definition	Value
Durations (months)		Productivity (US\$ per hour)	
Unemployed – Average duration to meet one:		Average:	
Job offer	6.217	Formal employees	3.437
Self-employed opportunity	37.037	Informal employees	2.402
Employed – Average duration to termination when:		Self-employed	1.038
Formal employee	20.408	Standard Deviation:	
Informal employee	17.857	Formal employees	0.892
Self-employed	37.037	Informal employees	0.864
Wage Offers (US\$ per hour)		Self-employed	0.506
Average:			
Formal employees	1.501		
Informal employees	1.307		
Standard Deviation:			
Formal employees	0.384		
Informal employees	0.432		

NOTE: Implied by the point estimates in the previous table.

- Informal salaried jobs are the **least stable** state; self-employment is by far the **most stable**.
- $\beta_s = 0.321$ — 22% of the minimum wage — exceeds $\beta_1 = 0.303$: self-employment is an amenity-bearing choice, not disguised unemployment. [Maloney (1999); Hamilton (2000)]

Experiments

Experiments: Definitions

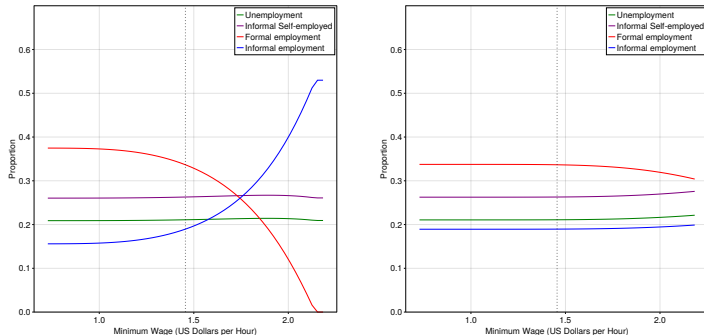
We vary the mandatory minimum wage from 50% below to 50% above its 2023 value. For each value we solve for the new equilibrium, simulate labor market careers under it, and compute statistics on the simulated sample.

Four versions, designed to isolate the two margins we claim matter:

1. **Endogenous Vacancy Creation** — the baseline. All agents adjust, including firms' posting.
2. **Exogenous Vacancy Creation** — arrival rates λ_1, λ_0 fixed at estimated values, so only workers reoptimize.
 - The environment assumed by most prior work. (1) vs (2) prices the demand-side margin.
3. **Endogenous Vacancy Creation, no Self-employment** — firms adjust, but self-employment is unavailable.
 - (1) vs (3) prices self-employment as a separate labor market state.
4. **Exogenous Vacancy Creation, no Self-employment** — both margins shut down.

Welfare throughout is U : the present discounted value of participating in this labor market. It prices the whole career, not the wage in any one state.

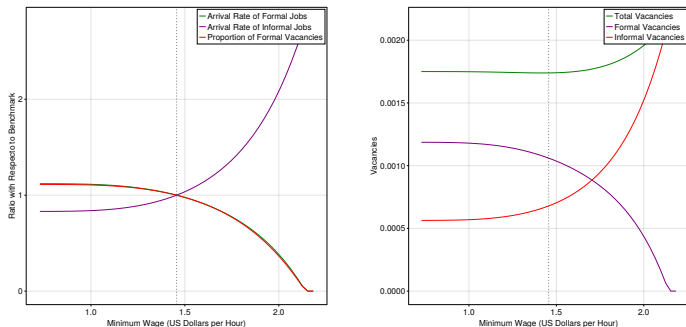
Results: Labor Market States



NOTE: Left — endogenous vacancy creation. Right — exogenous vacancy creation.

- Raising w_M **drops formal employment** and raises every other state, most of all informal salaried work: at about **US\$1.7 per hour, informal employees outnumber formal ones**.
- Changes are **asymmetric**: far more elastic to increases than to decreases.
- With vacancy posting fixed (right), the sensitivity is **orders of magnitude smaller**.

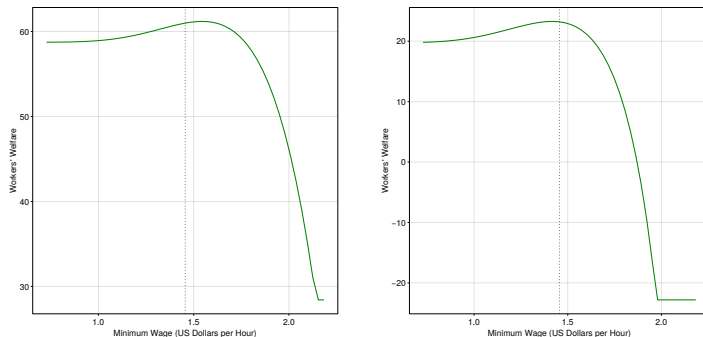
Results: Firms Substitute Informal for Formal Vacancies



NOTE: Endogenous vacancy creation. Left — arrival rates of offers to workers. Right — composition of posted vacancies.

- A higher w_M cuts the expected profit of a filled formal job, so ν^* falls: firms **substitute informal for formal postings** and informal offers arrive faster.
- Reservation values move the same way — fewer formal offers **and** a higher bar to accept one
wages & reservation values .
- Yet average informal wages are **stable over a long stretch**, falling only once the formal wage distribution approaches a mass point at w_M spike & profits .

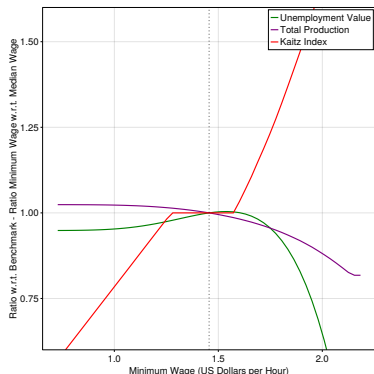
Results: Workers' Welfare



NOTE: Both under endogenous vacancy creation. Left — baseline. Right — no self-employment.

- Welfare is **non-monotonic**: rising up to about **US\$1.55 per hour** — slightly *above* the current value — then declining rapidly.
- So the welfare-maximizing minimum wage is **positive and close to current levels**.
- Without self-employment (right), the peak arrives **slightly below** the current value: a stable, amenity-bearing state **cushions** the adjustment.

Results: Welfare, Production and the Kaitz Index



- A binding minimum wage **reduces total production even where it raises welfare**: between about US\$1.00 and US\$1.55 per hour welfare rises while production falls mildly. Beyond that, both fall.
- The Kaitz index (w_M over the median employee wage) explains why. It rises linearly **up to one**: at the benchmark most employees are formal and about half of them earn exactly w_M , so the median sits at the floor.
- Over that range, raising w_M **widens $[\underline{x}_M, \bar{x}_M]$** , where surplus passes from firms to workers. Eventually the **employment effect** dominates and welfare falls.
- Welfare and output respond **differently**: optimal design is a genuine trade-off.

Conclusion

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- The specification needs only standard household-survey moments plus a vacancy count, so it is **portable across LAC**. Estimates for further countries are in progress.

Appendix

APPENDIX

- Stationary, continuous time; infinitely lived agents discount at ρ ; **linear utility**.
- Four labor market states: unemployment u , formal salaried e_1 , informal salaried e_0 , self-employed s .
- **Search frictions**: CRS matching function $Q(u, v)$, tightness $\theta = v/u$, $v = \sum_{f=0,1} v_f$:
 - unemployed meet an employer of type f at rate $\lambda_f = \pi_f \lambda(\theta)$, where $\pi_f = v_f / (v_1 + v_0)$;
 - firms meet workers at rate $\varphi(\theta) = q(\theta)/\theta$ — all firms compete in the **same** market [Acemoglu (2001)];
 - self-employment opportunities arrive at rate γ , with net income $y \sim R(y)$.
- On meeting, both sides observe match productivity $x \sim G_f(x)$ and formality f , then **bargain** over w and decide whether to accept.
- Termination shocks η_f (salaried) and δ (self-employment) return the worker to unemployment, so **both** formal→informal and informal→formal transitions occur.
- The **institutional context** differs by state:
 - **formal salaried**: payroll contribution t , benefit flow β_1 , mandatory minimum wage w_M ;
 - **informal salaried**: no t , no w_M , but a flow punishment cost c paid by the firm;
 - **self-employed**: no t , no c , no w_M , plus an amenity flow β_s ;
 - **unemployed**: flow value β_u , net of search costs.

Model: Firms and the Posting Decision

This is where the **demand side** becomes an equilibrium object rather than a parameter.

- Posting a vacancy costs κ ; posting a *formal* one costs an **additional firm-specific** $\nu \sim F(\nu)$.
- Free entry drives the value of both types of vacancy to zero, $V_f(\nu) = 0$:

$$\frac{\kappa + f \times \nu}{\varphi} = \int \max \left\{ \frac{x - w_f(x) - ftw_f(x) - (1-f)c}{\rho + \eta_f}, 0 \right\} dG_f(x); \quad f = 1, 0$$

- A firm posts formal only if ν is covered by expected formal profit, defining a **reservation cost** ν^* :

$$\nu^* = \varphi \int \max \left\{ \frac{x - (1+t)w_1(x)}{\rho + \eta_1}, 0 \right\} dG_1(x) - \kappa$$

- So the **formal share of vacancies is endogenous**:

$$\pi_1 = \Pr[\nu \leq \nu^*] = F(\nu^*)$$

- A policy that cuts the expected profit of formal jobs lowers ν^* and π_1 : it **reallocates the offers workers receive**.

Present discounted values in recursive form — flow value plus continuation value:

$$(\rho + \lambda + \gamma)U = \beta_u + \lambda \sum_{f=0,1} \pi_f \int \max[E_f[w_f(x)], U] dG_f(x) + \gamma \int \max[S(y), U] dR(y)$$

$$(\rho + \delta)S[y] = y + \beta_s + \delta U$$

$$(\rho + \eta_f)E_f[w_f(x)] = w_f(x) + f \times \beta_1 + \eta_f U$$

$$(\rho + \eta_f)J_f[x] = x - w_f(x) - ftw_f(x) - (1 - f)c + \eta_f V_f$$

- λ depends on aggregate tightness θ and the π_f depend on the number of vacancies created: the **supply and demand sides are linked through $\lambda_f = \pi_f \lambda$** .
- ρU is the **present discounted value of participating in this labor market**. It is the worker's outside option in every bargain — and it is our **welfare measure** in the counterfactuals.

Model: Wage Determination

Wages solve the Nash bargaining problem, with worker bargaining weight α :

$$\begin{aligned}w_1(x) &= \operatorname{argmax}[w_1(x) + \beta_1 - \rho U]^\alpha [x - (1+t)w_1(x)]^{(1-\alpha)} \text{ s.t. } w_1(x) \geq w_M \\w_0(x) &= \operatorname{argmax}[w_0(x) - \rho U]^\alpha [x - w_0(x) - c]^{(1-\alpha)}\end{aligned}$$

Interior solutions map productivity one-to-one into wages:

$$w_0(x) = \alpha(x - c) + (1 - \alpha)\rho U \qquad w_1(x) = \alpha \frac{x}{1+t} + (1 - \alpha)[\rho U - \beta_1]$$

- Bargaining **passes institutional costs to workers**: the firm pays t and c , but both lower the wage at a given x .
- Both schedules depend on ρU — the channel through which a minimum wage covering **only formal jobs** still moves **informal** wages.
- Monotonicity delivers the reservation productivities:

$$x_0^* = \rho U + c \qquad x_1^* = (1+t)[\rho U - \beta_1] \qquad y^* = \rho U - \beta_s$$

Model: The Minimum Wage and the Spike

Definition (Binding Minimum Wage)

A mandatory minimum wage w_M is binding if and only if $w_M > w_1^*$.

If binding, a **corner solution** appears alongside the interior one, delimited by two cutoffs:

$$\bar{x}_M = \frac{(1+t)}{\alpha} \{w_M - (1-\alpha)[\rho U - \beta_1]\} \qquad \underline{x}_M = (1+t)w_M$$

- Just below $\bar{x}_M$ the bargained wage would fall below w_M . The parties cannot agree on it, but the match still yields surplus — so they deviate and pay **exactly** w_M .
- Optimal down to $\underline{x}_M$, where firm profits are zero; below it the match is rejected.

$$w_1(x) = \begin{cases} w_M & \text{if } \underline{x}_M \leq x < \bar{x}_M \\ \alpha \frac{x}{1+t} + (1-\alpha)[\rho U - \beta_1] & \text{otherwise} \end{cases}$$

- Over $[\underline{x}_M, \bar{x}_M)$ surplus is **transferred from firms to workers** relative to the Nash split, and raising w_M **widens the interval** — how a minimum wage can *raise* welfare.

Definition (Equilibrium)

Given:

1. The primitive parameters $\{\rho, \eta_f, \gamma, \delta, \alpha, \kappa\}$, $f = 0, 1$ and the matching function $Q(u, v)$;
2. The distributions $G_f(x)$, $R(y)$, and $F(\nu)$;
3. The institutional parameters $\{t, w_M, c\}$ and utility parameters $\{\beta_u, \beta_1, \beta_s\}$;

the *steady state equilibrium* is a value U satisfying the Bellman equation for unemployment, a market tightness θ satisfying the free entry condition for informal vacancies, and a proportion of formal vacancies π_1 determined by the free entry condition for formal vacancies — together with the invariant distribution of individuals across labor market states.

- **Endogenous** in the model: the accepted wage distributions, employment and informality levels, the distribution over labor market states, market tightness, and the **formal/informal composition of vacancies**.

The Likelihood Function

Back

Hazard out of unemployment, evaluated at the reservation values:

$$\zeta_u = \lambda_1 \tilde{G}_1((1+t) \max\{\rho U - \beta_1\}, w_M) + \lambda_0 \tilde{G}_0(\rho U + c) + \gamma \tilde{R}(\rho U - \beta_s)$$

Probability that a formal employee is paid exactly the minimum wage:

$$\Pr[w_j = w_M | e_1] = \frac{\tilde{G}_1((1+t)w_M) - \tilde{G}_1\left(\frac{(1+t)}{\alpha}\{w_M + (1-\alpha)[\rho U - \beta_1]\}\right)}{\tilde{G}_1((1+t)w_M)}$$

Complete log-likelihood:

$$\begin{aligned} \ln L(\Theta; t, w, y) = & \sum_{j \in u} \ln f_{j,u}(t_{j,u}, j \in u) + \sum_{j \in e_1} I[w_j = w_M] \ln \Pr[w_j = w_M, t_{j,e_1}, j \in e_1] \\ & + \sum_{j \in e_1} I[w_j > w_M] \ln f_{j,e_1}(w_j, w_j > \max\{\rho U - \beta_1\}, t_{j,e_1}, j \in e_1) \\ & + \sum_{j \in e_0} \ln f_{j,e_0}(w_j, w_j \geq \rho U, t_{j,e_0}, j \in e_0) + \sum_{j \in s} \ln f_{j,s}(y_j, y_j \geq \rho U - \beta_s, t_{j,s}, j \in s) \end{aligned}$$

State probabilities solve the steady-state flow balance:

$$\lambda_1 \tilde{G}_1((1+t) \max\{\rho U - \beta_1\}, w_M)u = \eta_1 e_1; \quad \lambda_0 \tilde{G}_0(\rho U + c)u = \eta_0 e_0; \quad \gamma \tilde{R}(\rho U - \beta_s)u = \delta s$$

Recovering the Demand-Side Parameters

Back

- Formal share of vacancies, from the estimated arrival rates:

$$\pi_1 = \frac{\lambda_1}{\lambda_1 + \lambda_0}$$

- Matching elasticity, assuming a one-parameter Cobb-Douglas $Q(v, u) = v^\phi u^{1-\phi}$ so that $\lambda = \theta^\phi$:

$$\phi = \frac{\log \lambda_1 - \log \pi_1}{\log v - \log u}$$

- Vacancy posting cost, from free entry for **informal** vacancies (with $\varphi = \theta^{\phi-1}$):

$$\kappa = \theta^{\phi-1} (1 - \alpha) \int_{\rho U + c} \left[\frac{x - c - \rho U}{\rho + \eta_0} \right] dG_0(x)$$

- Distribution of the extra cost of a formal posting, assuming ν exponential, $F(\nu) = 1 - \exp(-\psi\nu)$:

$$\psi = \frac{-\log(\pi_1 - 1)}{\nu^*}$$

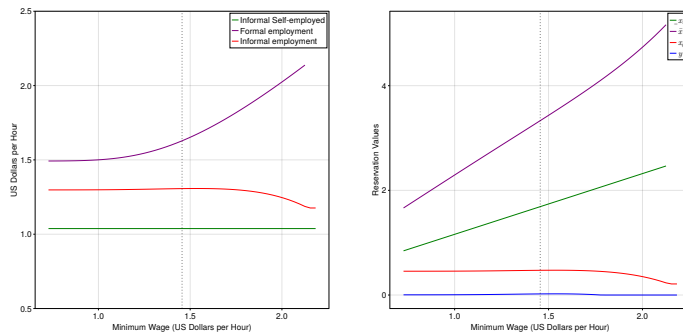
κ and $\kappa + \nu$ come out very high — multiples of the average productivity of a match — implied by the very high rate at which firms meet workers (about once every two days). Other estimates in this literature are also high, though an order of magnitude less so.

Table 3: Model Fit

Statistic	Model	Data	Statistic	Model	Data
Labor Market States Proportions			Mean and SD Wages (cont.)		
u	0.211	0.254	$E[w e_1]$	1.627	1.628
s	0.263	0.253	$SD[w e_1]$	0.275	0.276
e_1	0.337	0.317	$E[w e_1, w > w_M]$	1.801	1.810
e_0	0.190	0.176	$SD[w e_1, w > w_M]$	0.304	0.302
$P[w = w_M e_1]$	0.500	0.497	$E[w e_0]$	1.311	1.302
Mean and SD Wages			$SD[w e_0]$	0.438	0.366
$E[w s]$	1.041	1.079	Mean Durations		
$SD[w s]$	0.511	0.460	$E[t u]$	6.217	5.125
			$E[t s]$	37.426	37.426
			$E[t e_1]$	20.555	20.555
			$E[t e_0]$	18.006	18.006

NOTE: *Model* moments obtained by simulating 50,000 individuals at the point estimates. *Data* moments from the 2023 GEIH estimation sample.

Figure 2: Average Wages and Reservation Productivities (Endogenous Vacancy Creation)

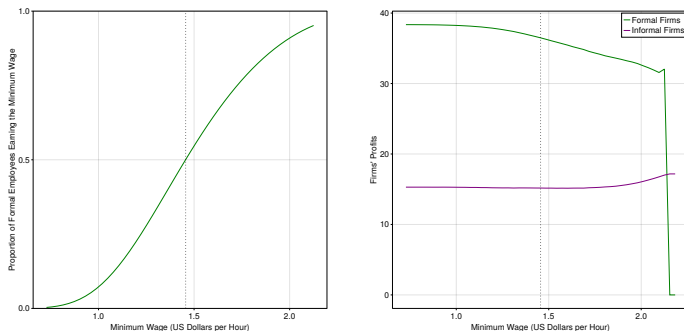


NOTE: Left — average wages and self-employment income by state. Right — reservation productivities. Average informal wages fall only once w_M approaches US\$2 per hour; self-employment income barely responds.

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Figure 3: Formal Employees at the Minimum Wage, and Present Value of Profits



NOTE: Both under endogenous vacancy creation. Left — proportion of formal employees earning exactly w_M , i.e. the mass over $[\underline{x}_M, \bar{x}_M)$. Right — average present discounted value of profits.